Research Agent: write-up

Rishabh Singh | Xiarch Agentic AI Engineer Intern take-home | github.com/imrishabh1999/xiarch-agent

What I built

An agent that takes a goal like *"Find the top 3 developments in solid-state batteries from the last week"*, plans the work, searches live news, checks every result, fixes its own mistakes, and writes a report where every finding links to its source. It uses three tools: web search, a calculator and a file writer. A **--mock** flag runs everything with saved data, so it works without an API key or internet, and an **--inject** flag makes any tool fail on purpose.

How it plans

The LLM (Groq, gpt-oss-120b) turns the goal into 3-7 steps, each with a tool, its inputs and a reason, and lists the guesses it made about the goal. Then plain Python code checks the plan: real tools, correct inputs, steps in the right order. If the plan is wrong, the LLM gets one more try with the exact problems listed. If the goal can't be understood at all, the agent asks a question instead of guessing. The plan is printed before anything runs, so the planning is visible.

How it handles things going wrong

My main design choice was to let the LLM suggest and let code check. Every result goes through a critic written in plain code, so it can't repeat the LLM's mistakes. It catches errors, empty or broken output (even when the tool says "ok"), news older than 7 days, findings that cite sources that don't exist, and numbers that aren't in the source. Each problem maps to a fix in a small table: try again with a short wait, search with new words, show the LLM its mistake and ask again, do one wider search, use saved results, or make a new plan once. Hard limits (20 tool calls, 3 tries per step, 4 minutes) stop endless loops. If nothing works, it still writes a report that says "failed" and makes nothing up. If it can only back up 2 of 3 findings, it reports 2 and says why.

How I tested it

46 unit tests cover each part. I also built an 8-goal check that runs real goals from start to end: a normal run, a different topic, each kind of injected failure, a full search outage, a vague goal, and a topic with almost no news. "Pass" means the agent did the right thing. In an outage the right thing is an honest failed report, and inventing findings for an empty topic counts as a fail. Final live result: 8 out of 8.

A bug the testing found

The first live check scored 6/8. Two battery runs found only 2 findings, right after the same goal had found 3. The step-by-step log showed the critic rejecting all results as older than 7 days. The cause was in the search library: it quietly switches between news sites, and its Bing code reads a date like "Sep 25" as "25 days ago", so fresh news looked old. I switched the search to DuckDuckGo with Yahoo as a backup, recorded which site each result came from, and added a test for it. The next check scored 8/8. Without the log and the 8-goal check, this would have looked like the LLM randomly doing worse.

What I would do next

Read full articles instead of just snippets; add a check for wrong claims written in words, not just numbers; score how trustworthy each news site is; and stop earlier on topics with no news, which currently use most of the tool budget before giving up.

I used Claude (an AI assistant) to help draft and review the code. I ran, tested and debugged every step myself and can explain all of it.